

Section 01

Industrial Security (Bachelor)

Section 01 – Introduction to Industrial Security

Introduction to Industrial Security

Lecture Notes

Department of Computer Science

- 1 Cyber-Physical Foundations
- 2 Context and History
- 3 Threat Setting and Misconceptions

By the end of this section you will be able to

- Define Operational Technology (OT) and contrast it with Information Technology (IT).
- Identify the core components of an Industrial Control System (ICS).
- Explain why the CIA triad is re-ordered for industrial environments.
- Place industrial security in its historical and economic context.
- Recognise common misconceptions about ICS security.

1 Cyber-Physical Foundations

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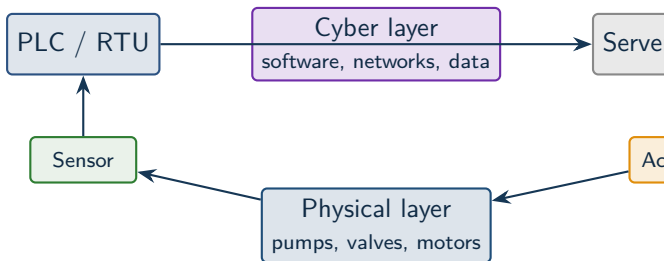
What is Industrial Security?

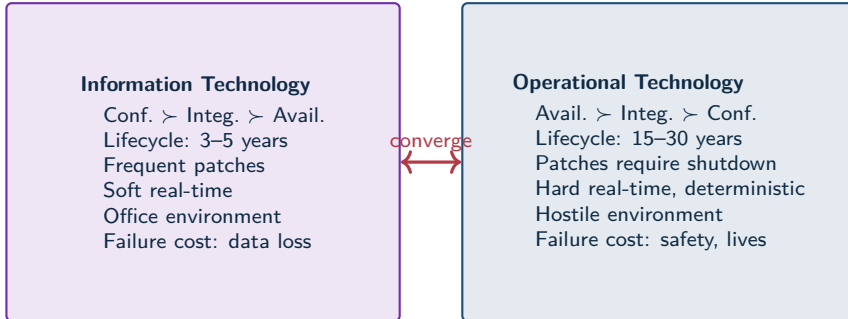
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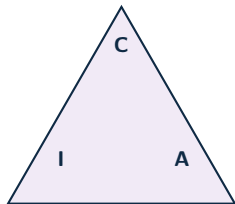
Industrial security protects **cyber-physical systems**: computers that monitor and control machines in the physical world.

Sectors: power, water, oil & gas, manufacturing, transport, building automation.

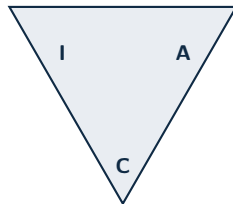
A breach has consequences in atoms, not just in bits.







IT priorities



OT priorities

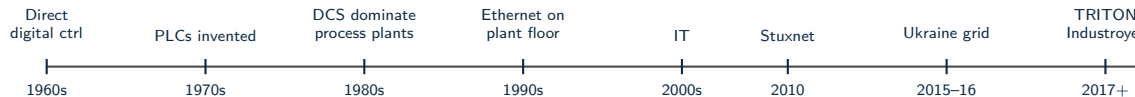
In OT, **availability** and **safety** dominate. A blocked alarm can blind an operator to a runaway reactor.

2 Context and History

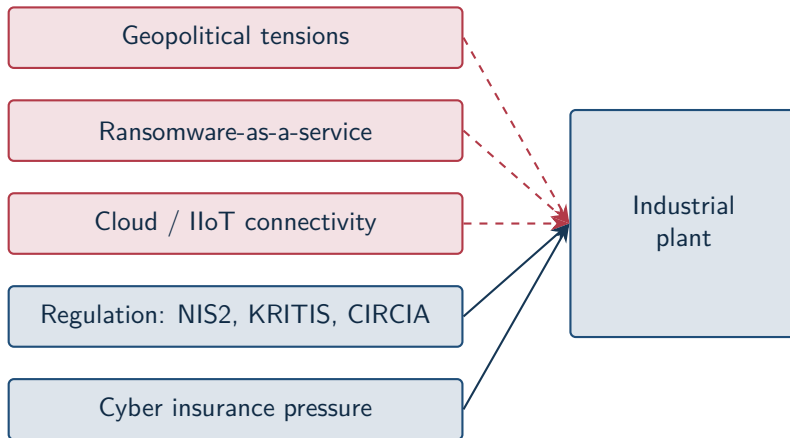
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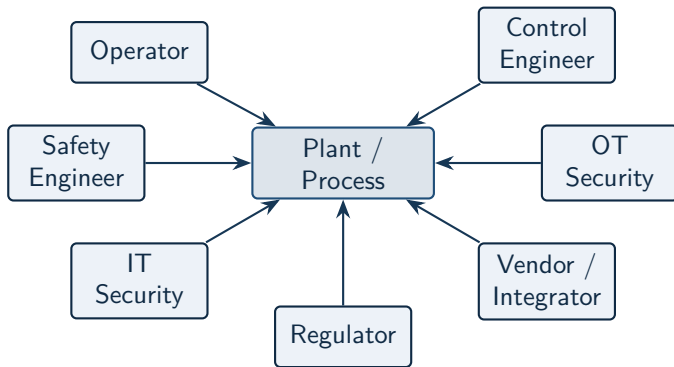
A Brief History of Industrial Control

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Each milestone made systems more capable – and more reachable.





Industrial security is a team sport across roles that rarely sat together historically.

3 Threat Setting and Misc

"We're air-gapped, so we're safe."

"Our protocols are obscure; nobody understands them."

"We've never been attacked, so we're fine."

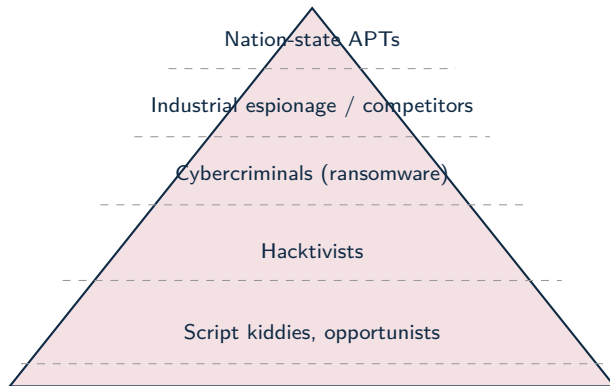
"IT tools will solve OT problems."

"Patching solves everything."

"Compliance equals security."

Threat Pyramid – Who Targets ICS?

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Capability rises as we move up; volume of actors falls.

OT differs fundamentally from IT in goals, lifecycle, and tolerance for failure.

Availability and safety dominate the CIA triad in industrial contexts.

Adversaries today span from script kiddies to nation-state APTs.

The next sections build the toolbox: components, networks, protocols, defenses.

End of Section 01



Introduction to Industrial Security

Questions and discussion